US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12406418
Kind Code B2
Date of Patent September 02, 2025
Inventor(s) Shi; Jing et al.

Personalized text-to-image generation

Abstract

One or more aspects of a method, apparatus, and non-transitory computer readable medium include obtaining an input description and an input image depicting a subject, encoding the input description using a text encoder of an image generation model to obtain a text embedding, and encoding the input image using a subject encoder of the image generation model to obtain a subject embedding. A guidance embedding is generated by combining the subject embedding and the text embedding, and then an output image is generated based on the guidance embedding using a diffusion model of the image generation model. The output image depicts aspects of the subject and the input description.

Inventors: **Shi; Jing (San Jose, CA), Xiong; Wei (San Jose, CA), Lin; Zhe (Clyde Hill, WA), Jung; Hyun Joon (Monte Sereno, CA)**

Applicant: **ADOBE INC.** (San Jose, CA)

Family ID: **93121742**

Assignee: **ADOBE INC. (San Jose, CA)**

Appl. No.: **18/476504**

Filed: **September 28, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240355022 A1	Oct. 24, 2024

Related U.S. Application Data

us-provisional-application US 63497353 20230420

Publication Classification

Int. Cl.: **G06T11/60** (20060101); **G06T7/194** (20170101); **G06T9/00** (20060101)

U.S. Cl.:

CPC **G06T11/60** (20130101); **G06T7/194** (20170101); **G06T9/00** (20130101); G06T2207/20081 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11922550	12/2023	Ramesh	N/A	G06F 40/284
2019/0279074	12/2018	Lin	N/A	G06F 16/583
2020/0151222	12/2019	Mannar	N/A	G06F 16/951
2020/0250537	12/2019	Li	N/A	G06F 18/22
2020/0311542	12/2019	Wang	N/A	G06N 5/02
2020/0380027	12/2019	Aggarwal	N/A	G06F 16/538
2022/0036127	12/2021	Lin	N/A	G06T 11/00
2022/0130499	12/2021	Zhou	N/A	G06V 10/25
2023/0237709	12/2022	Xu	345/625	G06V 10/774
2023/0245418	12/2022	Zhang	382/156	G06F 16/954

OTHER PUBLICATIONS

Gal, et al., An Image is Worth OneWord: Personalizing Text-to-Image Generation using Textual Inversion, arXiv preprint arXiv:2208.01618v1 [cs.CV] Aug. 2, 2022, 26 pages. cited by applicant

Ruiz, et al., “DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation”, arXiv preprint arXiv:2208.12242v2 [cs.CV] Mar. 15, 2023, 25 pages. cited by applicant

Kumari, et al., “Multi-Concept Customization of Text-to-Image Diffusion”, arXiv preprint arXiv:2212.04488v2 [cs.CV] Jun. 20, 2023, 26 pages. cited by applicant

Balaji, et al., “eDiff-I: Text-to-Image Diffusion Models with an Ensemble of Expert Denoisers” arXiv preprint arXiv:2211.01324v5 [cs.CV] Mar. 14, 2023, 24 pages. cited by applicant

Rombach, et al., “High-Resolution Image Synthesis with Latent Diffusion Models”, arXiv preprint arXiv:2112.10752v2 [cs.CV] Apr. 13, 2022, 45 pages. cited by applicant

Vonynov, et al., “P+: Extended Textual Conditioning in Text-to-Image Generation”, arXiv preprint arXiv:2303.09522v3 [cs.CV] Jul. 15, 2023, 16 pages. cited by applicant

Zhang, et al., “Adding Conditional Control to Text-to-Image Diffusion Models”, arXiv preprint arXiv:2302.05543v1 [cs.CV] Feb. 10, 2023, 33 pages. cited by applicant

Li, et al., “GLIGEN: Open-Set Grounded Text-to-Image Generation”, arXiv preprint arXiv:2301.07093v2 [cs.CV] Apr. 17, 2023, 21 pages. cited by applicant

Dong, et al., “DreamArtist: Towards Controllable One-Shot Text-to-Image Generation via Positive-Negative Prompt-Tuning”, arXiv preprint arXiv:2211.11337v3 [cs.CV] Apr. 5, 2023, 10 pages. cited by applicant

Gal, et al., “Encoder-based Domain Tuning for Fast Personalization of Text-to-Image Models”, arXiv preprint arXiv:2302.12228v3 [cs.CV] Mar. 5, 2023, 11 pages. cited by applicant
Han, et al., “SVDiff: Compact Parameter Space for Diffusion Fine-Tuning”, arXiv preprint arXiv:2303.11305v4 [cs.CV] Jul. 2, 2023, 22 pages. cited by applicant
Chen, et al., “Subject-driven Text-to-Image Generation via Apprenticeship Learning”, arXiv preprint arXiv:2304.00186v4 [cs.CV] Jun. 2, 2023, 17 pages. cited by applicant
Wei, et al., “ELITE: Encoding Visual Concepts into Textual Embeddings for Customized Text-to-Image Generation”, arXiv preprint arXiv:2302.13848v1 [cs.CV] Feb. 27, 2023, 10 pages. cited by applicant
Ma, et al., “Unified Multi-Modal Latent Diffusion for Joint Subject and Text Conditional Image Generation”, arXiv preprint arXiv:2303.09319v1 [cs.CV] Mar. 16, 2023, 14 pages. cited by applicant

---

Primary Examiner: Pringle-Parker; Jason A

Attorney, Agent or Firm: F. CHAU & ASSOCIATES, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to, and the benefit of, U.S. Provisional Application Ser. No. 63/497,353 filed on Apr. 4, 2023, entitled PERSONALIZED TEXT-TO-IMAGE GENERATION. The entire contents of the foregoing application are hereby incorporated by reference for all purposes.

BACKGROUND
(1) The present disclosure relates to image generation using a machine learning model. Image generation is a field of machine learning that relates to generating novel image content. Models used for image generation may include auto-encoders, generate adversarial networks (GANs), and diffusion models. Diffusion models utilize a noise map and a denoising operation to generate images. The diffusion model can include a forward process that adds noise to the input data (e.g., digital image) through a serious. The forward process can be followed by a reverse process that reconstructs an image by denoising the data.
SUMMARY
(2) Embodiments of the present disclosure provide a machine learning model including a generative network that can learn a concept from a set of images, then generate new scenes or styles of the concept from an input prompt. Personalized image synthesis may generate new images of a particular subject (e.g., person, animal, object, etc.) with different poses, backgrounds, locations, positions, orientations, dressing, lighting, styles, all while keeping the same subject's identity.
(3) A method, apparatus, and non-transitory computer readable medium for generating an image are described. One or more aspects of the method, apparatus, and non-transitory computer readable medium include obtaining an input description and an input image depicting a subject. One or more aspects of the method, apparatus, and non-transitory computer readable medium further include encoding the input description using a text encoder of an image generation model to obtain a text embedding, and encoding the input image using a subject encoder of the image generation model to obtain a subject embedding. One or more aspects of the method, apparatus, and non-transitory computer readable medium further include generating a guidance embedding by combining the subject embedding and the text embedding, and generating an output image based on the guidance embedding using a diffusion model of the image generation model, wherein the output image

depicts the subject and the input description.
(4) A method, apparatus, and non-transitory computer readable medium for training an image generation model are described. One or more aspects of the method, apparatus, and non-transitory computer readable medium include obtaining a training data set including a training image. One or more aspects of the method, apparatus, and non-transitory computer readable medium further include training the image generation model including a subject encoder and a diffusion model based on the training set, wherein the subject encoder is trained to encode an input image depicting a subject to obtain a subject embedding, and wherein the diffusion model is trained to generate an output image depicting the subject based on the subject embedding.
(5) A method, apparatus, and non-transitory computer readable medium for generating an image are described. One or more aspects of the method, apparatus, and non-transitory computer readable medium include one or more processors, and one or more memories including instructions executable by the one or more processors. One or more aspects of the method, apparatus, and non-transitory computer readable medium further include an image generation model comprising parameters stored in the one or more memories, wherein the image generation model is configured to receive a plurality of images as input, and is trained to generate a new image based on a feature embedding and a text embedding generated from the plurality of images and an input description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an illustrative depiction of a high-level diagram of users interacting with an image generator system, including a neural network for generating a new image of a subject in a novel arrangement, according to aspects of the present disclosure.
(2) FIG. **2** shows a flow diagram for a method of generating a customized image using a machine learning model and a text description, according to aspects of the present disclosure.
(3) FIG. **3** shows a block diagram of an example of an image generator, according to aspects of the present disclosure.
(4) FIG. **4** shows a diagram of a personalized text-to-image generator, according to aspects of the present disclosure.
(5) FIG. **5** shows a block diagram of an example of a guided diffusion model, according to aspects of the present disclosure.
(6) FIG. **6** shows a diffusion process according to aspects of the present disclosure.
(7) FIG. **7** shows an example of a U-Net, according to aspects of the present disclosure.
(8) FIG. **8** shows a diagram of an example of a method of generating an image, according to aspects of the present disclosure.
(9) FIG. **9** shows a diagram of an example of a method of training an image generator, according to aspects of the present disclosure.
(10) FIG. **10** shows a diagram of an example of a method of generating an image, according to aspects of the present disclosure.
(11) FIG. **11** shows a diagram of an example of a method of training an image generation model, according to aspects of the present disclosure.
(12) FIG. **12** shows an example of a computing device for image generation, according to aspects of the present disclosure.
DETAILED DESCRIPTION
(13) The present disclosure relates to generating customized digital images. In some case, an image is generated based on an initial set of images of an individual subject (e.g., person, animal, building, object, etc.) and a descriptive text prompt, that changes aspects of the individual subject in the image, while maintaining sufficient detail to still recognize the subject in the new image.

(14) Image generations models can be fine-tuned to generate custom images in a particular style or displaying a particular concept. In some cases, a training task for custom image generation can be referred to as a recognition problem. A recognition problem can be seen as an instance-conditioned generation task with identity preservation and language control. Using a compact embedding to represent the concept can lead to a lack of fine-grained identity details in the generated images. Furthermore, fine-tuning an image generation model can be costly and time consuming.
(15) Various embodiments of the disclosure enable efficient generation of customized images by learning a concept from a set of representative inputs. An image generation apparatus can utilize a text-to-image model (e.g., diffusion model, variational autoencoder, etc.) with trainable adapter layers to extract rich identity information from the input images and introduce them into the fixed backbone of the pre-trained model. Using an adapter layer to learn a concept from a limited set of images can enable efficient, real-time fine tuning of a personalized image generation model.
(16) In various embodiments, by using an adapter having trainable adapter layers, the model can preserve the identity details of the input concept while keeping the generation ability and language controllability of the pre-trained model. Input images can be converted to a textual token for general subject learning, and adapter layers can be introduced to adopt rich image representation for generating fine-grained identity details.
(17) In various embodiments, new high-quality images can be generated from text description, p, and a few images of a subject (i.e., concept), using a pre-trained text-to-image model. A unique identifier, V, can be added to the input to represent the particular subject, and a learnable image encoder can be used to map the input images to a subject embedding. A pre-trained diffusion model can utilize the subject embedding along with the embedding of the original text prompts to generate new images of the input subject. The adapter layers can be added to the pre-trained model to take rich patch features extracted from the input images for better identity preservation. A denoising loss can be used to learn the new components, while the original weights of the pre-trained model are frozen. Training can involve recovering the input image through denoising guided by conditions.
(18) The subject embedding, however, may miss the fine details of the input subject, such as the shape of object parts, textural details, structure details, and so on. To better preserve the subject identity, rich features containing identity details can be introduced into the pre-trained diffusion model.
(19) In various embodiments, a machine learning model that can generate text-aligned, identity-preserved, and high-fidelity image variations of an input concept is provided.
(20) In various embodiments, the network architecture can be trained and used to generate a new image with the same identifiable subject as a set of input images, but with a different arrangement of object(s), location(s), position(s), attribute(s), material, expression, and style. Because the subject of the input concept in the images may not make up a large enough portion of the image, the subject may be cropped out from each input image to obtain a set of conditional images, which can force the machine learning model to focus on the exact object. The background of each cropped image may also be masked out, where the mask may be generated at the pixel level of the image. Images with multiple potential subjects/objects can be filter out to simplify the training.
(21) In various embodiments, rich patch feature tokens can be extracted from the input images with a patch encoder, and the rich patch feature tokens injected into adapter layers of a U-Net for greater identity preservation. The U-Net of a pre-trained diffusion model can use the prompt embeddings and the rich patch feature tokens as conditions to generate new images of the input subject. Vision Transformers (ViT) can take advantage of transformers by treating an image as a sequence of patch tokens (e.g., 16×16 patches).
(22) In various embodiments, {circumflex over (V)}, can be a unique identifier used to represent the input subject, and [class noun] is a coarse category of the subject. A text prompt can have a format of " . . . {circumflex over (V)} [class noun] . . . " For example, if the original prompt is "A photo of a person playing guitar", then the modified prompt with the identifier is "A photo of a

{circumflex over (V)} person playing guitar", where "person" is the class noun.
(23) Accordingly, embodiments of the disclosure improve on a personalized image generation process and system, where the machine learning model can achieve identity preservation even with one input image at inference time. The model can generate language-aligned, identity-preserved images on unseen concepts with only a single forward pass.
(24) Network Architecture
(25) One or more aspects of the apparatus and method include one or more processors; a memory coupled to and in communication with the one or more processors, wherein the memory includes instructions executable by the one or more processors to perform operations including:
(26) In various embodiments, an encoder architecture can be used to encode the images and learn a concept from a set of images, then generate new scenes or styles of the concept from an input prompt. The original weights of the pre-trained model can be frozen (i.e., fixed) and the model can be extended with new trainable adapter layers. In various embodiments, an original transformer block contains a self-attention layer followed by a cross-attention layer that takes both the visual feature tokens and the textual embeddings as inputs for cross-attention learning. In each transformer block, a new learnable adapter layer can be added between the two frozen layers. The new layer can be formulated as:
(27) $y^{'} := y +$ .Math. tanh( ) .Math. $S([y, f_p])$.
(28) FIG. **1** is an illustrative depiction of a high-level diagram of users interacting with an image generator system, including a neural network for generating a new image of a subject in a novel arrangement, according to aspects of the present disclosure.
(29) In various embodiments, an image generation system **100** can involve a user **105** who can interact with image generating software on a user device **110**. The user device **110** can communicate with an image generation apparatus **120**, which can be a server located on the cloud **130**. The image generation apparatus **120** can generate a new image **125** from an input set **112** of images **115**, where each of the images **115** contain the same subject. The image generation apparatus **120** can obtain the set **112** of images **115** from the user device **110** or from a database **140**, as instructed by the user **105**.
(30) Embodiments of the disclosure can be implemented in a server operating from the cloud **130**, where the cloud **130** is a computer network configured to provide on-demand availability of computer system resources, such as data storage and computing power. In some examples, the cloud **130** provides resources without active management by the user **105**. The term cloud is sometimes used to describe data centers available to many users over the Internet. Some large cloud networks have functions distributed over multiple locations from central servers. A server is designated an edge server if the server has a direct or close connection to a user. In some cases, a cloud **130** is limited to a single organization. In other examples, the cloud **130** is available to many organizations. In an example, a cloud **130** includes a multi-layer communications network comprising multiple edge routers and core routers. In another example, a cloud **130** is based on a local collection of switches in a single physical location.
(31) In various embodiments, the functions of the image generation apparatus **120** can be located on or performed by the user device **110**. User device **110** may be a personal computer, laptop computer, mainframe computer, palmtop computer, personal assistant, smart phone, tablet, mobile device, or any other suitable processing apparatus. In some non-limiting examples, user device **110** includes software that incorporates an image generation application. In some examples, the image generation application on user device **110** may include functions of image generation apparatus **120**. Images **115** and other resources for obtaining the images **115** may be stored on the user device **110**, a database **140**, or a combination thereof.
(32) In various embodiments, a user interface may enable user **105** to interact with user device **110**. In some embodiments, the user interface may include an audio device, such as an external speaker system, an external display device such as a display screen, or an input device (e.g., remote control

device interfaced with the user interface directly or through an I/O controller module). In some cases, a user interface may be a graphical user interface (GUI). In some examples, a user interface may be represented in code which is sent to the user device and rendered locally by a browser.
(33) In various embodiments, an image generation apparatus **120** can include a computer implemented network comprising a user interface, and a machine learning model, which can include a diffusion model. The image generation apparatus **120** can also include a processor unit, a memory unit, a training component, a mask component, a noise component, and an image generation component. The training component can be used to train the machine learning model. Additionally, image generation apparatus **120** can communicate with database **140** via cloud **130**. In some cases, the architecture of the image generation model is also referred to as a network model. The image generation component can be trained to generate a new image using the image generation model. Further detail regarding the architecture of image generation apparatus **120** is provided for example with reference to FIGS. **3-7**, and **10**. Further detail regarding the operation of image generating apparatus **120** is provided, for example, with reference to FIGS. **2-11**.
(34) In various embodiments, image generating apparatus **120** is implemented on a server. A server provides one or more functions to users linked by way of one or more communication networks. In some cases, the server can include a single microprocessor board, which includes a microprocessor responsible for controlling aspects of the server. In some cases, a server uses one or more microprocessors and protocols to exchange data with other devices/users on one or more of the communication networks via hypertext transfer protocol (HTTP), and simple mail transfer protocol (SMTP), although other protocols such as file transfer protocol (FTP), and simple network management protocol (SNMP) may also be used. In some cases, a server is configured to send and receive hypertext markup language (HTML) formatted files (e.g., for displaying web pages). In various embodiments, a server comprises a general-purpose computing device, a personal computer, a laptop computer, a mainframe computer, a supercomputer, or any other suitable processing apparatus.
(35) Cloud **130** can be a computer network configured to provide on-demand availability of computer system resources, such as data storage and computing power. In some examples, cloud **130** provides resources without active management by user **105**. The term “cloud” is sometimes used to describe data centers available to many users (e.g., user **105**) over the Internet. Some large cloud networks have functions distributed over multiple locations from central servers. A server is designated an edge server if the server has a direct or close connection to a user (e.g., user **105**). In some cases, cloud **130** is limited to a single organization. In other examples, cloud **130** is available to many organizations. In an example, cloud **130** includes a multi-layer communications network comprising multiple edge routers and core routers. In another example, cloud **130** is based on a local collection of switches in a single physical location.
(36) Database **140** is an organized collection of data, where for example, database **140** can store data in a specified format known as a schema. Database **140** may be structured as a single database, a distributed database, multiple distributed databases, or an emergency backup database. In some cases, a database controller may manage data storage and processing in database **140**. In some cases, a user **105** interacts with the database controller. In other cases, a database controller may operate automatically without user interaction.
(37) FIG. **2** shows a flow diagram for a method of generating a customized image using a machine learning model and a text description, according to aspects of the present disclosure.
(38) In various embodiments, given a few images of a subject, new high-quality images of this subject can be generated from a text description, p, where the generated image variations preserve the identity of the input subject. A representation of someone or something in images, that is a subject of the images, and a representation of a new portrayal of the subject in words can be received by the machine learning model, and a new, custom image showing the same subject in a novel arrangement not seen in the received images can be generated.

(39) At operation **210**, image generation apparatus **120** can prompt a user **105** to provide a plurality of initial images **115**, where the set **112** of the plurality of images **115** contains the same subject, which can be a person, an object, a building, an animal, etc., where the subject is the primary visual element that is the focus of the image **115**. The subject can be present on a background that presents a scene for the subject, where the subject and scene can present a concept. The image generation apparatus **120** can prompt a user to provide a textual description for a new image to be generated, where the requested textual description provides attributes and details about the subject and background different from the initial images obtained by the image generation apparatus **120**.
(40) At operation **220**, the user **105** can provide a set **112** of images **115** to the image generation apparatus **120**, where the images **115** may be provided from a user device **110** or identified on another device, for example, a database **140** in communication with the user device **110** and the image generation apparatus **120**. The images can be a set **112** of images **115** that each includes the same person, object, building, animal, etc., where the image generation apparatus **120** can use the set **112** of images **115** and a learnable image encoder to map the initial images **115** to a subject embedding.
(41) At operation **230**, the image generation apparatus **120** can obtain the set **112** of images **115** and the text description, where the images **115** can be obtained from the user device **110** or database **140**, and the text description can be obtained from the user **105**. The text description prompts can have a format of “ . . . {circumflex over (V)} [class noun] . . . ”, where {circumflex over (V)} is a unique identifier to represent the input subject, and [class noun] is a coarse category of the subject. A modified prompt, p.sub.s, can be obtained from the original; prompt, p, by putting the identifier right before the class noun of the subject.
(42) In various embodiments, a natural language processing (NLP) model can receive and analyze a text description prompt provided by a user **105**. The natural language processing (NLP) model can be trained and used to interpret the text description provided by the user **105**.
(43) At operation **240**, the image generation apparatus **120** can map the initial images **115** to a subject embedding. In some cases, the initial images **115** can be converted into a subject embedding by mapping the images to a compact concept feature vector, f.sub.c, in the textual latent space. For example, a subject image encoder, E.sub.c, can convert the input images to a compact textual embedding, f.sub.c. The subject embedding can represent identity attributes of a subject (i.e., attributes of the subject that are invariant when the subject is depicted in a different pose or scene, or that enable a subject to be recognized in different scenes or poses).
(44) In various embodiments, a pre-trained diffusion model (DM) takes the subject embedding along with the embedding of the original prompts (i.e., text description) to generate new images of the input subject. Contrastive Language-Image Pre-Training (CLIP) is an example of a multi-modal neural network trained on a variety of (image, text) pairs, where the network can be instructed in natural language to predict the most relevant text snippet, given an image, without directly optimizing for the task.
(45) In various embodiments, the final textual embeddings of the input prompt can be obtained from the Text embeddings, c.sub.s, of the modified prompt, e.g. c.sub.s=CLIP(p.sub.s). The embedding of identifier, {circumflex over (V)}, can be a word vector that can be replaced with the concept feature vector, f.sub.c, to obtain the subject injected textual embedding c. The learned subject embedding is a compact feature containing the global semantics of the input images.
(46) At operation **250**, rich patch features can be extracted from the images **115**, where the rich patch features can be used to capture the foreground detail and maintain the subject identity. More input images can provide more details of the foreground to better preserve the subject identity, where to better preserve the subject identity, rich features containing identity details can be injected into a pre-trained diffusion model.
(47) In various embodiments, rich patch feature tokens can be extracted from the input images **115** with a patch encoder.

(48) At operation **260**, a new image **125** can be generated based on the subject embedding, prompt embedding, and rich patch feature tokens, where the image can be generated by an image generation component, for example, a pre-trained diffusion model.
(49) In various embodiments, the pre-trained diffusion model can include adapter layers that can use rich patch feature tokens extracted from the input images for identity preservation, where the rich patch feature tokens can be injected into the adapter layers of a U-Net to improve identity preservation. An adapter can receive an arbitrary number of conditioning images as inputs during inference. In various embodiments, during training, the image encoders and the adapter layers are trainable, while the other parts are frozen.
(50) The U-Net of the pre-trained diffusion model takes the prompt embeddings and the rich patch feature tokens as conditions to generate the new images of the input subject.
(51) In various embodiments, the original weights of the pre-trained model can be frozen (i.e., fixed), and the model can be extended with new trainable adapter layers, where the pre-trained model retains the visual synthesis and language understanding abilities, but can further learn the additional visual features, as the rich patch feature tokens containing identity details. A denoising loss can be used to learn the new components, while the original weights of the pre-trained model are frozen.
(52) At operation **270**, a new image **125** can be provided to the user **105**, for example, by transmitting the new image over a communications network (e.g., Internet) to the user device **110**.
(53) FIG. **3** shows a block diagram of an example of an image generator, according to aspects of the present disclosure.
(54) In one or more embodiments, an image generator **300** obtains original image(s) **115** including original content, and receives a text description (e.g., prompt) indicating different content to be depicted in a new image. The original image(s) **115** can be stored in computer memory **320**.
(55) In various embodiments, the image generator **300** can include a computer system **380** including one or more processors **310**, computer memory **320**, a training component **330**, a mask component **340**, a noise component **350**, and an image generation model **360** (e.g., an image generation model). The computer system **380** of the image generator **300** can be operatively coupled to a display device **390** (e.g., computer screen) for presenting prompts and images to a user **105**, and operatively coupled to input devices to receive description input from the user. The image generation model **360** can be an image generation model, that can include a text encoder, a subject encoder, a patch encoder, and a diffusion model, as further described in reference to FIG. **4**.
(56) In various embodiments, the diffusion model can include a U-Net having an adapter layer. The text encoder can be, for example, a transformer, that receives a text prompt as input and generates a text embedding as output. The subject encoder can be, for example, a pre-trained multimodal encoder as the backbone followed by a randomly initialized fully-connected layer, that receives the images as input and generates a subject embedding as output. The patch encoder can be, for example, a pre-trained CLIP image encoder as the backbone followed by a randomly initialized fully-connected layer, that receives the images as input and generates a feature (i.e., patch) embedding as output. The diffusion model can be a stable diffusion model with a U-Net that can receive the text embedding, subject embedding, and feature embedding, and generates a new image.
(57) In various embodiments, training component **330** can receive a training data set for the image generation model **360**, and apply a loss function to results obtained from the model being trained using the training data set. The training component **330** can update the model weights of the image generation model **360** based on the results of the loss function.
(58) A description can include text indicating a desired edit to the original image(s) **115**, where the output image **125** is generated based on the description that describes the desired features of a subject and/or scene. In some aspects, a prompt from the image generator **300** includes an original image **115** presented to the user **105** on the display device **390** or communicated to a user's device

110.
(59) In various embodiments, mask component **340** identifies a mask indicating a region of the original image(s) to be removed. The mask component **340** can identify a region of the image to be ignored, so as to focus on the subject of the image.
(60) In various embodiments, noise component **350** generates a noise map based on the original image **115** and a mask, where the output image **125** is generated based on the noise map. In some examples, noise component **350** generates an iterative noise map for each of a set of output images with successively reduced noise to produce the output image.
(61) According to some aspects, image generation model **360** (e.g., diffusion model) generates an output image **125** including the original, identified subject from the original image(s) **115** and the modified content, where the new image **125** can be generated using a diffusion model that takes a vector generated from a description by a text encoder **430** (e.g., a transformer) as input. In some aspects, the output image **125** combines additional content in a manner consistent with the original content, such that the original subject is identifiable in the new image **125**. In some examples, a diffusion model iteratively produces a set of output images.
(62) Diffusion models are a class of generative models that convert Gaussian noise into images from a learned data distribution using an iterative denoising process. Diffusion models are also latent variable models with latent vectors, z={z.sub.t|t∈[0, 1]}, that obey a forward process q (z|x) starting at data x˜p(x). This forward process is a Gaussian process that satisfies the Markovian structure. For image generation, the diffusion model is trained to reverse the forward noising process (i.e., denoising, z.sub.t˜q(z.sub.t|x)). In addition, a text embedding from the natural language processor (NLP) can be used as a conditioning signal that guides the denoising process. A text encoder can encode the input text of the description into text embeddings, where the diffusion model maps the text embedding into an image.
(63) In various embodiments, the computation and parameters in a diffusion model take part in the learned mapping function which reduces noise at each timestep (denoted as F). The model takes as input x (i.e., noisy or partially denoised image depending on the timestep), the timestep t, and conditioning information the model was trained to use. In some cases, the conditioning information can be a text prompt (e.g., TP, “ ”, and AP are text prompts). Classifier-free guidance is a mechanism to vary and control the influence of the conditioning on the sampled distribution at inference. In some cases, the conditioning can be replaced by the null token (i.e., the empty string, “ ”, in case of text conditioning) during training. A single scalar can control effect of the conditioning during inference.
(64) Some examples of the method, apparatus, non-transitory computer readable medium, and system further include iteratively producing a plurality of output images. Some examples further include generating an iterative noise map for each of the plurality of output images with successively reduced noise to produce the output image.
(65) Embodiments of the disclosure utilize an artificial neural network (ANN), which is a hardware or a software component that includes a number of connected nodes (i.e., artificial neurons), which loosely correspond to the neurons in a human brain. Each connection, or edge, transmits a signal from one node to another (like the physical synapses in a brain). When a node receives a signal, the nodes process the signal and then transmits the processed signal to other connected nodes. In some cases, the signals between nodes comprise real numbers, and the output of each node is computed by a function of the sum of the node's inputs. In some examples, nodes may determine their output using other mathematical algorithms (e.g., selecting the max from the inputs as the output) or other suitable algorithms for activating the node. Each node and edge is associated with one or more node weights that determine how the signal is processed and transmitted.
(66) During the training process, these model weights are adjusted to improve the accuracy of the result (i.e., by minimizing a loss function which corresponds in some way to the difference between the current result and the target result). The weight of an edge increases or decreases the strength of

the signal transmitted between nodes. In some cases, nodes have a threshold below which a signal is not transmitted at all. In some examples, the nodes are aggregated into layers. Different layers perform different transformations on the layer's inputs. The initial layer is known as the input layer and the last layer is known as the output layer. In some cases, signals traverse certain layers multiple times.

(67) In various embodiments, a stable diffusion model can be used as the base generative model, and masked image synthesis method with stochastic differential equations may be used as a baseline. Note that the same hyperparameters (i.e., noise strength, total diffusion steps, sampling schedule, classifier free guidance strength C) can be used.

(68) FIG. **4** shows a diagram of a personalized text-to-image generator, according to aspects of the present disclosure.

(69) In various embodiments, an image generation model **360** in FIG. **3** can be, for example, image generation model **400** without test-time fine-tuning, where the model can be built upon a pre-trained text-to-image model. This involves, given a set **112** of images **115** of the same subject, for example, a person (e.g., man, woman, child), an animal (e.g., cat, dog, horse, etc.), an object, (e.g., car, plane, toaster, building, etc.) in a scene, as well as a language (i.e., text) prompt that describes how a new image of the input subject and scene should be constructed, the model can output a generated image with the same subject, while providing what the text prompt describes. The problem is difficult to solve because not only should the identity of the input subject be maintained, but the generated image should also follow the description in the text prompt. It is difficult to balance both subject identity and image description fidelity in the image generation model without test-time fine-tuning. For example, given input of several digital images (e.g., photos) of the same woman and a language (i.e., text) prompt, “a woman is eating at the Eiffel Tower,” the output image should be of the same woman eating at the Eiffel Tower.

(70) In various embodiments, subject encoder **410** and a patch encoder **420** can receive a set **112** of images **115**, wherein each of the images **115** in the set **112** includes a same subject (e.g., person, animal, object, etc.). The subject encoder **410** can learn the general concept of the input images **115** by converting the images **115** to a textual token. The subject encoder **410** can map the subject of the images **115** to the compact textual embedding. The patch encoder **420** can generate rich patch feature tokens **450** from the input images **115**, where the rich patch feature tokens **450** transform features of the image **115** to embeddings, f.sub.p.

(71) In various embodiments, images with multiple potential subjects/objects **116** can be filter out to simplify the training.

(72) In various embodiments, subject encoder **410** and a patch encoder **420** can share the same backbone **415** and have the last layers fully connected layers, but have different linear layers between the backbone and fully connected layers. The subject encoder **410** and the patch encoder **420** can each be a multi-modal encoder such as a CLIP image encoder. In some cases, the encoders have a shared backbone.

(73) In various embodiments, the subject image encoder **410**, E.sub.c can be used to map the images to a compact subject feature vector, f.sub.c, in the textual space, where f.sub.c is the average feature vector of the global features of all input images, given by:

(74) $f_c = .\mathrm{Math.}_{i=1}^{N} \frac{E_c(x_s^i)}{N}$.

(75) The averaging of a plurality of images with different backgrounds can provide filtering of the background portions of the images, while maintaining the subject identity. The use of more images at input with the same subject can improve removal of inconsistent features between images.

(76) In various embodiments, a text encoder **430** (e.g., a CLIP text encoder) can receive a text description **118**, as a prompt, and map the words to form a prompt (i.e., text) embedding **440**. A modified prompt, p.sub.s, can be obtained from the original prompt, p, by putting the identifier, {circumflex over (V)}, **445** right before the class noun **447** of the subject. For example, if the original prompt, p, is “A photo of a person playing guitar”, then the modified prompt, p.sub.s, with

the identifier, {circumflex over (V)}, is “A photo of a {circumflex over (V)} person playing guitar”, where “person” is the class noun. And for example, for “person” subject category, the identifier, {circumflex over (V)}, for the nouns that are coarse descriptions of “person”, including “man”, “woman”, “baby”, “girl”, “boy”, “lady”, etc. can be inserted. A frozen (i.e., fixed parameter) text encoder **430**, for example, a CLIP text encoder, can be used to map the other words, “A photo of,” to form the prompt embedding **440**. Other examples of a prompt can be, “a painting of {circumflex over (V)} [class noun] in Van Gogh style”, and “a photo of {circumflex over (V)} [class noun] in mountain with aurora.”
(77) In various embodiments, the input images can be mapped to a textual space, as a compact textual embedding, so that the models have a deep understanding of the concept (i.e., subject), but can lead to a lack of fine-grained identity details in the generated images. A few trainable adapter layers can be added to extract rich identity information from the input images and inject them into the fixed backbone of the pre-trained model. This can preserve the identity details of the input concept while keeping the generation ability and language controllability of the pre-trained model.
(78) In various embodiments, the embedding of identifier, {circumflex over (V)}, **445** can be replaced with the concept feature vector, f.sub.c, generated by the subject encoder **410**, to obtain the subject injected textual embedding c. This embedding can be the condition in the cross-attention layers **462** of a U-Net **460** in a text-to-image diffusion model. The rich patch feature tokens **450**, generated by the patch encoder **420**, can be introduced into the adapter layers **464** of a U-Net **460** for better identity preservation. In various embodiments, the U-Net **460** of the pre-trained diffusion model takes the prompt embeddings **440** with the concept feature vector, f.sub.c, and the rich patch feature tokens, f.sub.p, **450** as inputs to generate new images **470** of the subject in the images **115**. The data flow in FIG. **4** for the U-Net **460** is bottom-up from the self-attention layer **466** to the adapter layer **464** to the cross-attention layer **462**.
(79) During training, the subject image encoder **410**, the patch encoder **420**, and the adapter layers **464** are trainable, while the weights of the cross-attention layers **462** and self-attention layers **466** are fixed (i.e., frozen).
(80) CLIP is a multi-modal vision and language model, that can be used for image-text similarity and for zero-shot image classification. CLIP uses a ViT like transformer to get visual features and a causal language model to get the text features. Both the text and visual features are then projected to a latent space with identical dimensions. The dot product between the projected image and text features can be used as a similarity score.
(81) In a non-limiting exemplary embodiment, the image generation model **400** can utilize Stable Diffusion or a text-to-image transformer model, as the pretrained text-to-image model, where the model can be trained on text-image pairs. For both the concept encoder, E.sub.c, and the patch encoder, E.sub.p, the encoder can be used as the backbone followed by a randomly initialized fully-connected layer.
(82) FIG. **5** shows a block diagram of an example of a guided diffusion model, according to aspects of the present disclosure.
(83) The guided latent diffusion model **500** depicted in FIG. **5** is an example of, or includes aspects of, the corresponding diffusion model of the image generation model **360** described with reference to FIG. **3**.
(84) Diffusion models are a class of generative neural networks which can be trained to generate new data with features similar to features found in training data. In particular, diffusion models can be used to generate novel images. Diffusion models can be used for various image generation tasks including image super-resolution, generation of images with perceptual metrics, conditional generation (e.g., generation based on text guidance), image inpainting, and image manipulation.
(85) Types of diffusion models include Denoising Diffusion Probabilistic Models (DDPMs) and Denoising Diffusion Implicit Models (DDIMs). In DDPMs, the generative process includes reversing a stochastic Markov diffusion process. DDIMs, on the other hand, use a deterministic

process so that the same input results in the same output. Diffusion models may also be characterized by whether the noise is added to the image itself, or to image features generated by an encoder (i.e., latent diffusion).

(86) Diffusion models work by iteratively adding noise to the data during a forward process and then learning to recover the data by denoising the data during a reverse process. For example, during training, guided latent diffusion model **500** may take an original image **115** in a pixel space **505** as input and apply a forward diffusion process **510** to gradually add noise to the original image **115** to obtain noisy images **520** at various noise levels.

(87) Next, a reverse diffusion process **530** (e.g., a U-Net ANN) gradually removes the noise from the noisy images **520** at the various noise levels to obtain an output image **125**. In some cases, an output image **125** is created from each of the various noise levels. The output image can be compared to the original image to train the reverse diffusion process **530**.

(88) The reverse diffusion process **530** can also be guided based on a text prompt **540**, or another guidance prompt, such as a text description, an image, a layout, a segmentation map, etc. The text prompt **540** can be encoded using a text encoder **550** (e.g., a multimodal encoder) to obtain guidance features **560** in guidance space **570**. The guidance features **560** can be combined with the noisy images **520** at one or more layers of the reverse diffusion process **530** to ensure that the output image **125** includes content described by the text prompt **540**. For example, guidance features **560** can be combined with the noisy features using a cross-attention block within the reverse diffusion process **530**.

(89) FIG. **6** shows a diffusion process according to aspects of the present disclosure.

(90) According to embodiments of the present disclosure, the example includes forward diffusion process **605** and reverse diffusion process **610**. The example further includes noisy image **615**, first intermediate image **620**, second intermediate image **625**, and original image **630**. Original image **630** is an example of, or includes aspect of, the same element described with reference to FIGS. **1-2** and **4**.

(91) A diffusion model can include both a forward diffusion process **605** for adding noise to an image (or features in a latent space) and a reverse diffusion process **610** for denoising the images (or features) to obtain a denoised image. The forward diffusion process **605** can be represented as p(x.sub.t-1|x.sub.t), and the reverse diffusion process **610** can be represented as q(x.sub.t|x.sub.t-1). In some cases, the forward diffusion process **605** is used during training to generate images with successively greater noise, and a neural network is trained to perform the reverse diffusion process **610** (i.e., to successively remove the noise).

(92) In an example forward process for a latent diffusion model, the model maps an observed variable x.sub.0 (either in a pixel space or a latent space) to intermediate variables x.sub.1, . . . , x.sub.T using a Markov chain. The Markov chain gradually adds Gaussian noise to the data to obtain the approximate posterior q(x.sub.1:T|x.sub.0) as the latent variables are passed through a neural network such as a U-Net, where x.sub.1, . . . , x.sub.T have the same dimensionality as x.sub.0.

(93) The neural network (e.g., U-Net) may be trained to perform the reverse process. During the reverse diffusion process **610**, the model begins with noisy data x.sub.T, such as a noisy image **615** and denoises the data to obtain the p(x.sub.t-1|x.sub.t). At each step t−1, the reverse diffusion process **610** takes x.sub.t, such as first intermediate image **620**, and t as input. Here, t represents a step in the sequence of transitions associated with different noise levels. The reverse diffusion process **610** outputs x.sub.t-1, such as second intermediate image **625** iteratively until x.sub.T is reverted back to x.sub.0, the original image **630**. The reverse process can be represented as:

(94) $p \ (x_{t-1} \mid x_t) := \ (x_{t-1}; \quad (x_t, t),$ .Math. $(x_t, t))$. (1)

(95) The joint probability of a sequence of samples in the Markov chain can be written as a product of conditionals and the marginal probability:

(96) $x_T : p\ (x_{0:T}) := p(x_T)$ .Math.$_{t=1}^{T}$ $p\ (x_{t-1} \mid x_t)$, (2) where p(x.sub.T)=
custom character(x.sub.T; 0, I) is the pure noise distribution as the reverse process takes the outcome of the forward process, a sample of pure noise, as input and Π.sub.t=1.sup.T p.sub.θ(x.sub.t-1|x.sub.t) represents a sequence of Gaussian transitions corresponding to a sequence of addition of Gaussian noise to the sample.
(97) At interference time, observed data x.sub.0 in a pixel space can be mapped into a latent space as input and a generated data {tilde over (x)} is mapped back into the pixel space from the latent space as output. In some examples, x.sub.0 represents an original input image with low image quality, latent variables x.sub.1, . . . , x.sub.T represent noisy images, and {tilde over (x)} represents the generated image with high image quality.
(98) FIG. **7** shows an example of a U-Net, according to aspects of the present disclosure.
(99) The U-Net **700** depicted in FIG. **7** is an example of, or includes aspects of, the architecture used within the reverse diffusion process described herein.
(100) In some examples, diffusion models are based on a neural network architecture known as a U-Net. The U-Net **700** takes input features **705** having an initial resolution and an initial number of channels, and processes the input features **705** using an initial neural network layer **710** (e.g., a convolutional network layer) to produce intermediate features **715**. The intermediate features **715** are then down-sampled using a down-sampling layer **720** such that down-sampled features **725** have a resolution less than the initial resolution and a number of channels greater than the initial number of channels.
(101) This process is repeated multiple times, and then the process is reversed. That is, the down-sampled features **725** are up-sampled using up-sampling process **730** to obtain up-sampled features **735**. The up-sampled features **735** can be combined with intermediate features **715** having a same resolution and number of channels via a skip connection **740**. These inputs are processed using a final neural network layer **745** to produce output features **750**. In some cases, the output features **750** have the same resolution as the initial resolution and the same number of channels as the initial number of channels.
(102) In some cases, U-Net **700** takes additional input features to produce conditionally generated output. For example, the additional input features could include a vector representation of an input prompt (e.g., text description). The additional input features can be combined with the intermediate features **715** within the neural network at one or more layers. For example, a cross-attention module can be used to combine the additional input features and the intermediate features **715**.
(103) Embodiments of the disclosure can utilize an artificial neural network (ANN), which is a hardware and/or a software component that includes a number of connected nodes (i.e., artificial neurons), which loosely correspond to the neurons in a human brain. Each connection, or edge, transmits a signal from one node to another (like the physical synapses in a brain). When a node receives a signal, the node processes the signal and then transmits the processed signal to other connected nodes. In some cases, the signals between nodes comprise real numbers, and the output of each node is computed by a function of the sum of the node's inputs. In some examples, nodes may determine their output using other mathematical algorithms (e.g., selecting the max from the inputs as the output) or other suitable algorithms for activating the node. Each node and edge is associated with one or more node weights that determine how the signal is processed and transmitted.
(104) In various embodiments, an NLP can be a transformer type natural language processor/neural language model (e.g., GPT), an encoder based natural language processor (e.g., Bidirectional Encoder Representations from Transformers (BERT), Robustly Optimized BERT (RoBERTa)), or other encoder/decoder based NLP.
(105) A transformer or transformer network is a type of neural network model used for natural language processing tasks. A transformer network transforms one sequence into another sequence

using an encoder and a decoder. Encoder and decoder include modules that can be stacked on top of each other multiple times. The modules comprise multi-head attention and feed forward layers. The inputs and outputs (target sentences) are first embedded into an n-dimensional space. Positional encoding of the different words (i.e., give every word/part in a sequence a relative position since the sequence depends on the order of its elements) are added to the embedded representation (n-dimensional vector) of each word. In some examples, a transformer network includes attention mechanism, where the attention looks at an input sequence and decides at each step which other parts of the sequence are important.
(106) The attention mechanism involves query, keys, and values denoted by Q, K, and V, respectively. Q is a matrix that contains the query (vector representation of one word in the sequence), K represents all the keys (vector representations of all the words in the sequence), and V is the values, which is the vector representations of all the words in the sequence. For the encoder and decoder, multi-head attention modules, V consists of the same word sequence as Q. However, for the attention module that is taking into account the encoder and the decoder sequences, V is different from the sequence represented by Q. In some cases, values in V are multiplied and summed with some attention-weights, a.
(107) Inference
(108) During the model's inference, the background of the cropped images can be masked out, but augmentations to the masked images may not be performed, i.e., custom character=None. Although the rich patch feature, f.sub.p, is composed of tokens from only one image during training, the adapter layers **464** can actually take an arbitrary number of conditioning images as inputs during inference. For example, a user could provide a single image **115** for input or a dozen images **115** for input. This flexibility can be attributed to the concatenation operation in the adapter layer and the nature of self-attention.
(109) FIG. **8** shows a diagram of an example of a method of generating an image, according to aspects of the present disclosure.
(110) At operation **810**, a set of one or more images of a subject are obtained, and provided to the image generation apparatus **120**, where the images **115** contain the same subject. An adapter can take an arbitrary number of conditioning images as inputs during inference due to the concatenation operation in the adapter layer and the nature of self-attention.
(111) At operation **820**, a text description for a new image to be generated is obtained, and provided to the image generation apparatus **120**, where the text description provides attributes and details to be applied to the new image. The input description provides a new depiction of the subject present in the plurality of input images.
(112) At operation **830**, the subject can be cropped from the images to enforce focus. The one or more images can be cropped to isolate the subject in each image. The training images contain the subject categories and the text description can contain the related coarse category nouns.
(113) At operation **840**, the backgrounds of the cropped images are masked out, where:
(114) $x_s^i := x_s^i$ .Math. m_s^i; where m.sub.s.sup.i is the mask for the subject in the i-th cropped image with 1 indicating the pixel of object and 0 indicating a background pixel. The entity segmentation masks can be collected for all the collected images using an entity segmentation model. The masked image set X.sub.s can be the final image condition for training the model.
(115) In various embodiments, augmentations may not be performed to the masked images, or random augmentations can be performed to the masked image for training, where augmentation can randomly remove portions of the training images. This can act as a pseudo-pair generation of multiple images having different portions dropped out from a single training image
(116) At operation **850**, the value of balance factor, β, is reduced below 1 during inference, so that the adapter layer takes both the visual information from the original pre-trained model and the conditioning images. A balance factor, β, can play the primary role for achieving a good balance between language understanding and identity preservation. Setting β to 1 for training can result in a

strong reconstruction of the input images with good identity preservation, while the language-image alignment is weakened. In various embodiments, the balance factor, β, can be about 0.1 to about 0.9, or about 0.2 to about 0.5, or about 0.3 during inference, although other ranges and values are also contemplated.
(117) At operation **860**, the subject token is renormalized with a factor of α∈(0, 1], which is equivalent to rescaling the cross-attention between the concept token and the visual tokens in the Cross-Attention layers. Renormalization can help to achieve a better balance between text-image alignment and identity preservation. With renormalization, the attentions of the nouns are more balanced, and the model successfully generates the style and the background of the prompt description.
(118) To address this issue, the concept token can be renormalized with a factor of α∈(0, 1], where:
(119) f_c = .Math. f_c .
(120) The testing results indicate that larger β or α can both contribute to better identity preservation but weaker language comprehension ability, where values of β=0.3, α=0.4 can be used as a trade-off. Since there are only linear mappings from the prompt embeddings before calculating the cross-attention, such renormalization strategy is essentially equivalent to rescaling the cross-attention between the concept token and the visual tokens in the Cross-Attention layers **462**, such that the attention of the concept token does not dominate the cross-attention. With renormalization, the attentions of the nouns are more balanced, and the model successfully applied the text description without loss of the subject identity.
(121) At operation **870**, a new image can be generated by an image generation component based on the input images and the text description, where the balance factor, β, and the renormalization factor, a, can achieve balance between text-image alignment and identity preservation.
(122) Although the rich patch feature, f.sub.p, is composed of tokens from only one image during training, the adapter can take an arbitrary number of conditioning images as inputs during inference. This flexibility is owning to the concatenation operation in the adapter layer and the nature of self-attention.
(123) X.sub.t={x.sub.t.sup.i}.sub.1.sup.N represents the set of the original images, where N is the number of input images. Because the portion of the input image showing the subject may not be large enough in comparison to the image size and/or background, the subject can be cropped out from each image to obtain a set of conditional images, X.sub.s={x.sub.s.sup.i}.sub.1.sup.N. Images that have the subject occupying less than 10% of the image area or greater than 70% of the image area may be filtered out (e.g., removed) from the set entirely. To further force the model to focus on the particular subject, the background of each cropped image can also be masked out, where:
(124) $x_s^i := x_s^i$ .Math. m_s^i; where m.sub.s.sup.i is the mask for the subject in the i-th cropped image with 1 indicating the pixel of object and 0 indicating a background pixel.
(125) In various embodiments, during training, random augmentations can be performed to the masked image, where x.sub.s.sup.i:=custom character(x.sub.s.sup.i);
(126) where custom character is the augmentation operator. The masked image set X.sub.s can be the final image condition for training the model.
(127) f_c = .Math.$_{i=1}^{N}$ $\frac{E_c(x_s^i)}{N}$.
(128) The new layer is formulated as:
(129) $y' := y +$ .Math. tanh( ) .Math. $S([y, f_p])$, where S is the self-attention operator, y, is the visual feature tokens, γ is a learnable scalar initialized as zero, the balance factor, β, is a constant to balance the weight and importance of the adapter layer, f.sub.p represents the rich patch token sequences extracted using an image encoder, E.sub.p, from input images X.sub.s. The self-attention operator, y, is the output from the of the self-attention layers **466** that is fed into the adjoining adapter layer **464**. The learnable scalar, γ, is the learnable portion of the adapter layers **464**, and represents an activation map. The output, y′, is the output of the adapter layer **464**, where the

adapter layers **464** are trainable self-attention layers.
(130) In a non-limiting exemplary embodiment, for each image **115**, **257** visual tokens are extracted with the patch encoder, E.sub.p, and the rich patch feature vector, f.sub.p, is obtained as a token sequence by concatenating the tokens from all the input images, where:
(131) $0f_p = \text{Concat}(\{E_p(x_s^i)\}_1^N)$; where x.sub.s.sup.i are the conditional images from 1 to N.
(132) Compared to the compact subject embedding, the patch feature contains subject-related content details of the input images; and thus can improve identity preservation. Rich patch features, as represented by a token sequence, can be extracted from images for the identity preservation.
(133) In various embodiments, only the adapter layers and the fully-connected layer of the image encoders are updated. The subject of the input training images can be learned by the subject image encoder **410** in FIG. **4**, and mapped to a compact textual embedding. The original weights of the pre-trained text-to-image model are frozen in the full training procedure. The subject encoder **410** and a patch encoder **420** are trainable. The model can be optimized with only the denoising loss of the diffusion model, where the object masks of the input images can be omitted for simplicity.
(134) In various embodiments, the model can be tested without cropping or masking the subject in the images. Identity similarity of the subject can be measured with the similarity of visual features between the input image and the generated image. A reconstruction score can be calculated between each pair of a newly generated image and the input images to evaluate identity preservation. An L2 norm can be used on the embeddings of the newly generated image and the input image to calculate a distance between the two subjects, where a lower distance indicates a greater similarity between the subjects.
(135) Model Training
(136) In various embodiments, the original image set, X.sub.t, without cropping out the object region or masking out the background, is regarded as the ground-truth. During training, heavy augmentation, custom character, can be used to obtain variations of masked images, X.sub.s. The training data set may not include paired images of the same subject, so one (1) image may be used to train the model per subject, where N=1 in the training data set. The model can be separately trained for each category.
(137) In various embodiments, during training, for each subject category, the corresponding nouns in the original prompt can be detected, and an identifier inserted. For example, for "person" subject category, the identifier for the nouns that are coarse descriptions of "person", including "man", "woman", "baby", "girl", "boy", "lady", etc. can be inserted. For a "cat" category, the identifier to the nouns, "cat", "kitten" and so on, can be inserted.
(138) In various embodiments, the subject can be encoded into the model, where the input images can be converted into a textual subject embedding. The identifier, {circumflex over (V)}, indicates the location of the textual embedding. A subject image encoder E.sub.c can be used to map the images to a compact subject feature vector, f.sub.c, in the textual space, where f.sub.c is the average feature vector of the global features of all input images. The embedding of identifier, {circumflex over (V)}, can be replaced with the subject feature vector, f.sub.c, to obtain the subject injected textual embedding c. This final embedding can be the condition in the cross-attention layers of the text-to-image diffusion model.
(139) In various embodiments, the loss function used to optimize the model is formulated as:
(140) $\mathcal{L} = \mathbb{E}_{z,t,cX_s,\eta\epsilon\ (0,1)}[\ .\text{Math.}\ \ -\ \ (z_t, t, c, X_s)\ .\text{Math.}_2^2]$; where z.sub.t is the latent noisy image at time step t obtained from the ground-truth image, x.sub.t.sup.1, η is the latent noise to predict, c is the textual embedding, X.sub.s is the set of conditioning images, and η.sub.θ is the noise prediction model with parameters θ. The model can be optimized with only the denoising loss of the diffusion model.
(141) In various embodiments, a single denoising step can be used for training, where the training loss can be calculated, as a loss function for an image at a single noise level.

(142) In various embodiments, an image can be generated by obtaining a plurality of input images containing the same subject and an input description of a new depiction of the subject, generating a rich patch feature token and a prompt embedding from the input images and the input description, and generating a new image based on the rich patch feature token and the prompt embedding using an image generation component.
(143) FIG. **9** shows a diagram of an example of a method of training an image generator, according to aspects of the present disclosure.
(144) At operation **910**, a plurality of images can be obtained for a data set, where each training image contains a subject, and a ground truth text description is associated with the image. Each image in the training data set can have a different subject.
(145) In various embodiments, images with multiple objects can be filtered out to simplify the training. Images where the region (e.g., area) ratio of the subject portion belonging to the target subject category is, for example, less than 0.1 or larger than 0.7 of the image area can be filtered out.
(146) At operation **920**, the background of the training images can be masked out to increase focus on the subject. During training of a model, the masks are applied to suppress the background content; therefore, the model learns to primarily keep the identity of the foreground object, but not the background.
(147) At operation **930**, augmentations can be performed to the training images, where the augmentations can be random. The augmentations can be to obtain variations of masked images X.sub.s, where random regional features replaced with a template randomly selected from a templates data set. The masked image set, X.sub.s, can be the final image condition to train the model.
(148) At operation **940**, the balance factor can be set to one (1) to have a strong reconstruction of the input images with good identity preservation.
(149) At operation **950**, noise can be generated by a noise component and added to a training image.
(150) At operation **960**, the noise added to the test image can be compared to a ground truth noise map for the training image to obtain a similarity value.
(151) At operation **965**, a loss value can be calculated for the comparison of the ground truth and the predicted noise.
(152) At operation **970**, the model parameters can be updated to improve the identity preservation ability and vision-language alignment of the model. To evaluate whether the identity can be fully preserved using a default prompt, a reconstruction can be used to measure the similarity of the visual features between the input image and the generated image.
(153) In various embodiments, an image generation model can be trained by obtaining a training data set including a plurality of training images and a text description, generating a new image based on an input test image and the text description using the image generation component, comparing the predicted noise and the ground truth noise, and updating parameters of the image generating component based on the comparison.
(154) In a non-limiting exemplary embodiment, the model is trained for 320 k iterations for person and 200 k iterations for cat, with the learning rate 1e-6 for adapter layers and 1e-4 for the FC layers in the visual encoders, under batch size 16 deployed over 4 A100 GPUs.
(155) Embodiments can utilize a word embedding model to encode a text prompt. A word embedding is a learned representation for text where words that have the same meaning have a similar representation. Glove, Word2Vec, GPT, BERT, and CLIP are examples of systems for obtaining a vector representation of words. GloVe is an unsupervised algorithm for training a network using aggregated global word-word co-occurrence statistics from a corpus. Similarly, a Word2vec model may include a shallow neural network trained to reconstruct the linguistic context of words. GloVe and Word2vec models may take a large corpus of text and produce a vector space

as output. In some cases, the vector space may have a large number of dimensions. Each word in the corpus is assigned a vector in the vector space. Word vectors are positioned in the vector space in a manner such that similar words are located nearby in the vector space. In some cases, an embedding space may include syntactic or context information in additional to semantic information for individual words.
(156) FIG. **10** shows a diagram of an example of a method of generating an image, according to aspects of the present disclosure.
(157) At operation **1010**, an input image is obtained, where the input image can be obtained from a user. The input image can be obtained by an image generation model, where the image generation model can include a text encoder, a subject encoder, a patch encoder, and a diffusion model. The diffusion model can include a U-Net having an adapter layer.
(158) At operation **1020**, an input description is obtained, where the input description can be a text description of a new image to be generated based on the input image using the image generation model. The input description can be obtained from the user.
(159) At operation **1030**, the input image is encoded to obtain a subject embedding, where the subject embedding can be generated by the subject encoder.
(160) At operation **1040**, the input description is encoded to obtain a text embedding, where the text embedding can be generated by the text encoder.
(161) At operation **1050**, a guidance embedding is generated, where the guidance embedding can be generated based on the subject embedding and the text embedding. The guidance embedding can be generated by combining the subject embedding and the text embedding, where the subject embedding can replace an identifier. The guidance embedding can be a subject injected textual embedding.
(162) At operation **1060**, an output image can be generated based on the guidance embedding, where the output image depicts the subject of the input image. The output image can be generated based on the guidance embedding using an image generation model, where the output image depicts the subject and the input description.
(163) At operation **1070**, the output image can be provided to the user, where the output image can identifiably depict the subject of the input image in a new arrangement described by the text description.
(164) In various embodiments, the input image can be encoded to obtain a feature embedding, where the feature embedding can be generated by the patch encoder. The feature embedding can be the embeddings, f.sub.p, based on the rich patch feature tokens **450** of the image(s) **115**.
(165) FIG. **11** shows a diagram of an example of a method of training an image generation model, according to aspects of the present disclosure.
(166) At operation **1110**, a training data set including a training image depicting a subject can be obtained. The training image can be obtained by an image generation model, where the image generation model can include a text encoder, a subject encoder, a patch encoder, and a diffusion model. The diffusion model can include a U-Net having an adapter layer.
(167) At operation **1120**, the training image can be encoded to obtain a subject embedding using a subject encoder.
(168) At operation **1130**, a noise prediction can be generated based on the subject embedding using the image generation model. The noise prediction can be generated using a diffusion model.
(169) At operation **1140**, the subject encoder can be trained based on the noise prediction.
(170) FIG. **12** shows an example of a computing device for image generation, according to aspects of the present disclosure.
(171) In various embodiments, the computing device **1200** includes processor(s) **1210**, memory subsystem **1220**, communication interface **1230**, I/O interface **1240**, user interface component(s) **1250**, and channel **1260**.
(172) In various embodiments, computing device **1200** is an example of, or includes aspects of

image generation apparatus **120**. In some embodiments, computing device **1200** includes one or more processors **1210** that can execute instructions stored in memory subsystem **1220** for generating a new image of an identifiable subject from an input image and a text description.

(173) In various embodiments, computing device **1200** includes one or more processors **1210**. In various embodiments, a processor **1210** can be an intelligent hardware device, (e.g., a general-purpose processing component, a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), a microcontroller, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or a combination thereof. In some cases, a processor **1210** is configured to operate a memory array using a memory controller. In other cases, a memory controller is integrated into a processor. In some cases, a processor is configured to execute computer-readable instructions stored in a memory to perform various functions. In some embodiments, a processor **1210** includes special-purpose components for modem processing, baseband processing, digital signal processing, or transmission processing.

(174) A processor **1210** may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Thus, the functions described herein may be implemented in hardware or software and may be executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor **1210**, the functions may be stored in the form of instructions or code on a computer-readable medium.

(175) In various embodiments, memory subsystem **1220** includes one or more memory devices. Examples of a memory device include random access memory (RAM), read-only memory (ROM), or a hard disk. Examples of memory devices **1220** include solid state memory and a hard disk drive. In some examples, memory is used to store computer-readable, computer-executable software including instructions that, when executed, cause a processor **1010** to perform various functions described herein. In some cases, the memory contains, among other things, a basic input/output system (BIOS) which controls basic hardware or software operation such as the interaction with peripheral components or devices. In some cases, a memory controller operates memory cells. For example, the memory controller can include a row decoder, column decoder, or both. In some cases, memory cells within a memory store information in the form of a logical state.

(176) According to some aspects, communication interface **1230** operates at a boundary between communicating entities (such as computing device **1200**, one or more user devices, a cloud, and one or more databases) and channel **1260** (e.g., bus), and can record and process communications. In some cases, communication interface **1230** is provided to enable a processing system coupled to a transceiver (e.g., a transmitter and/or a receiver). In some examples, the transceiver is configured to transmit (or send) and receive signals for a communications device via an antenna.

(177) According to some aspects, I/O interface **1240** is controlled by an I/O controller to manage input and output signals for computing device **1200**. In some cases, I/O interface **1240** manages peripherals not integrated into computing device **1200**. In some cases, I/O interface **1240** represents a physical connection or a port to an external peripheral. In some cases, the I/O controller uses an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or other known operating system. In some cases, the I/O controller represents or interacts with a user interface component, including, but not limited to, a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller is implemented as a component of a processor. In some cases, a user interacts with a device via I/O interface **1240** or via hardware components controlled by the I/O controller.

(178) According to some aspects, user interface component(s) **1250** enable a user to interact with computing device **1200**. In some cases, user interface component(s) **1250** include an audio device, such as an external speaker system, an external display device such as a display device **390** (e.g.,

screen), an input device (e.g., a remote-control device interfaced with a user interface directly or through the I/O controller), or a combination thereof. In some cases, user interface component(s) **1250** include a GUI.

(179) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of code or data. A non-transitory storage medium may be any available medium that can be accessed by a computer. For example, non-transitory computer-readable media can comprise random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), compact disk (CD) or other optical disk storage, magnetic disk storage, or any other non-transitory medium for carrying or storing data or code.

(180) Also, connecting components may be properly termed computer-readable media. For example, if code or data is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technology such as infrared, radio, or microwave signals, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technology are included in the definition of medium. Combinations of media are also included within the scope of computer-readable media.

(181) The description and drawings described herein represent example configurations and do not represent all the implementations within the scope of the claims. For example, the operations and steps may be rearranged, combined or otherwise modified. Also, structures and devices may be represented in the form of block diagrams to represent the relationship between components and avoid obscuring the described concepts. Similar components or features may have the same name but may have different reference numbers corresponding to different figures.

(182) Some modifications to the disclosure may be readily apparent to those skilled in the art, and the principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

(183) In this disclosure and the following claims, the word “or” indicates an inclusive list such that, for example, the list of X, Y, or Z means X or Y or Z or XY or XZ or YZ or XYZ. Also, the phrase “based on” is not used to represent a closed set of conditions. For example, a step that is described as “based on condition A” may be based on both condition A and condition B. In other words, the phrase “based on” shall be construed to mean “based at least in part on.” Also, the words “a” or “an” indicate “at least one.”

Claims

1. A method of generating an image, comprising: obtaining an input description and an input image depicting a subject; encoding the input description using a text encoder of an image generation model to obtain a text embedding; encoding the input image using a subject encoder of the image generation model to obtain a subject embedding; generating a guidance embedding by combining the subject embedding and the text embedding; and generating an output image based on the guidance embedding using a diffusion model of the image generation model, wherein the output image depicts one or more aspects of the input image and the input description.

2. The method of claim 1, wherein: the guidance embedding is generated by replacing an identifier in the text embedding with the subject embedding.

3. The method of claim 1, further comprising: encoding the input image to obtain a feature embedding representing a subject identity of the input image, wherein the output image is generated based on the feature embedding.

4. The method of claim 3, further comprising: introducing the feature embedding into an adapter layer of the image generation model to preserve the subject identity in the output image.

5. The method of claim 1, wherein: the text embedding is generated using a first multi-modal encoder and the subject embedding is generated using a second multi-modal encoder.
6. The method of claim 1, further comprising: applying a balance factor and a renormalization factor to the subject embedding.
7. The method of claim 6, wherein: the balance factor is less than 1.